Module 2:

1) Develop an Android application implementing AppBar (Toolbar) and Options Menu.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:background="#FFFFFF">

   <androidx.appcompat.widget.Toolbar
      android:id="@+id/toolbar"
      android:layout_width="match_parent"
      android:layout_height="?attr/actionBarSize"
      android:background="#1565C0"
      android:elevation="4dp"
      android:theme="@style/ThemeOverlay.AppCompat.Dark.ActionBar"
      android:title="My Application"
      android:titleTextColor="#FFFFFF" />

   <TextView
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:text="Welcome to My Application"
      android:textSize="24sp"
      android:textStyle="bold"
      android:textColor="#333333"
      android:gravity="center" />

</LinearLayout>
```

res/menu/options_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

   <item
      android:id="@+id/menu_home"
      android:title="Home"
      android:showAsAction="never" />

   <item
      android:id="@+id/menu_settings"
      android:title="Settings"
```

```
        android:showAsAction="never" />

    <item
        android:id="@+id/menu_about"
        android:title="About"
        android:showAsAction="never" />

</menu>
```

Output:

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2) Develop an Android application demonstrating AlertDialog and custom DialogFragment.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:background="#FFFFFF">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Dialog Demonstration"
        android:textSize="26sp"
        android:textStyle="bold"
        android:textColor="#1565C0"
        android:gravity="center"
        android:padding="20dp" />

    <Button
        android:id="@+id/btnAlertDialog"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Show AlertDialog"
        android:layout_margin="10dp" />

    <Button
        android:id="@+id/btnCustomDialog"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Show Custom Dialog"
```

```
    android:layout_margin="10dp" />

  <TextView
    android:id="@+id/dialogMessage"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Welcome to Android Application!"
    android:textSize="18sp"
    android:gravity="center"
    android:padding="20dp" />

</LinearLayout>
```

Output:

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3) Develop an Android application implementing SQLite database CRUD operations.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="#FFFFFF"
  android:padding="16dp">

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical">

    <TextView
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Student Database"
      android:textSize="26sp"
      android:textStyle="bold"
      android:textColor="#1565C0"
      android:gravity="center"
      android:padding="15dp" />

    <EditText
      android:id="@+id/rollNo"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Roll Number"
```

```
    android:inputType="number"
    android:padding="12dp" />

<EditText
    android:id="@+id/studentName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Student Name"
    android:inputType="textPersonName"
    android:padding="12dp" />

<EditText
    android:id="@+id/studentMarks"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Marks"
    android:inputType="numberDecimal"
    android:padding="12dp" />

<Button
    android:id="@+id/btnInsert"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Insert Student" />

<Button
    android:id="@+id/btnView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="View Students" />

<Button
    android:id="@+id/btnUpdate"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Update Student" />

<Button
    android:id="@+id/btnDelete"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Delete Student" />

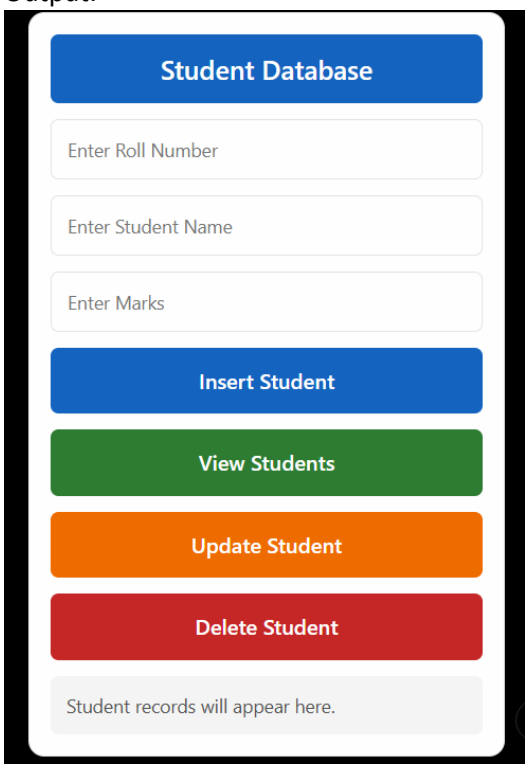
<TextView
    android:id="@+id/result"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Student records will appear here."
```

```
        android:textSize="16sp"
        android:padding="15dp"
        android:textColor="#333333" />

    </LinearLayout>

</ScrollView>
```

Output:



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4) Develop an Android application integrating SQLite with ListView or RecyclerView.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:background="#FFFFFF">

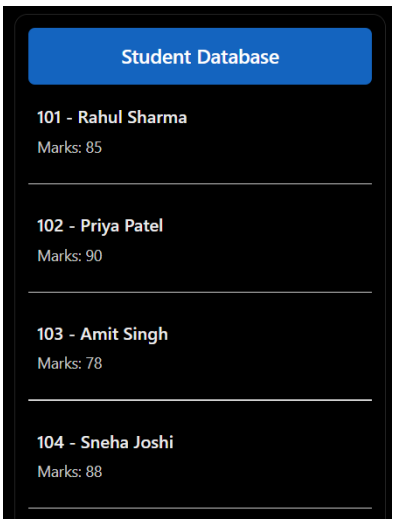
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Student Database"
```

```
        android:textSize="26sp"
        android:textStyle="bold"
        android:textColor="#1565C0"
        android:gravity="center"
        android:padding="15dp" />

    <ListView
        android:id="@+id/studentList"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:divider="#BDBDBD"
        android:dividerHeight="1dp"
        android:padding="8dp" />

</LinearLayout>
```

Output:



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5) Develop an Android application demonstrating Threads and Handlers for background processing.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:background="#FFFFFF">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
        android:text="Threads and Handlers"
        android:textSize="26sp"
        android:textStyle="bold"
        android:textColor="#1565C0"
        android:gravity="center"
        android:padding="20dp" />

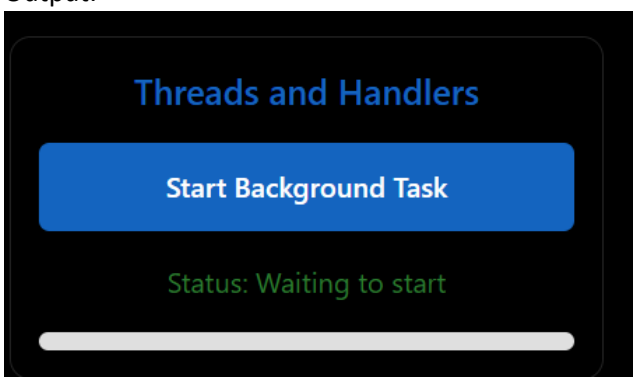
    <Button
        android:id="@+id/btnStart"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Start Background Task"
        android:textSize="18sp"
        android:layout_margin="10dp" />

    <TextView
        android:id="@+id/txtStatus"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Status: Waiting to start"
        android:textSize="20sp"
        android:textColor="#2E7D32"
        android:gravity="center"
        android:padding="20dp" />

    <ProgressBar
        android:id="@+id/progressBar"
        style="?android:attr/progressBarStyleHorizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:max="100"
        android:progress="0"
        android:visibility="visible"
        android:layout_margin="10dp" />

</LinearLayout>
```

Output:



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6) Develop an Android application implementing Static and Dynamic Broadcast Receivers.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp"
   android:background="#FFFFFF">

   <TextView
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Broadcast Receiver Demo"
      android:textSize="26sp"
      android:textStyle="bold"
      android:textColor="#1565C0"
      android:gravity="center"
      android:padding="20dp" />

   <Button
      android:id="@+id/btnSendBroadcast"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Send Broadcast"
      android:textSize="18sp"
      android:layout_margin="10dp" />

   <TextView
      android:id="@+id/txtStatus"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Status: Waiting for Broadcast"
      android:textSize="18sp"
      android:textColor="#2E7D32"
      android:gravity="center"
      android:padding="20dp" />

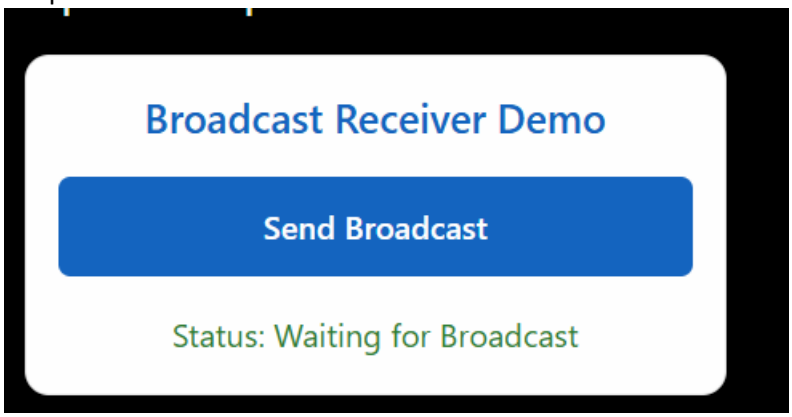
</LinearLayout>
```

AndroidManifest.xml

```
<receiver
   android:name=".MyStaticReceiver"
   android:exported="false">
```

```
    <intent-filter>
        <action android:name="com.example.MY_BROADCAST" />
    </intent-filter>
</receiver>
```

Output:



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